

React JS Hands-On Lab

Event Handling – eventexamplesapp

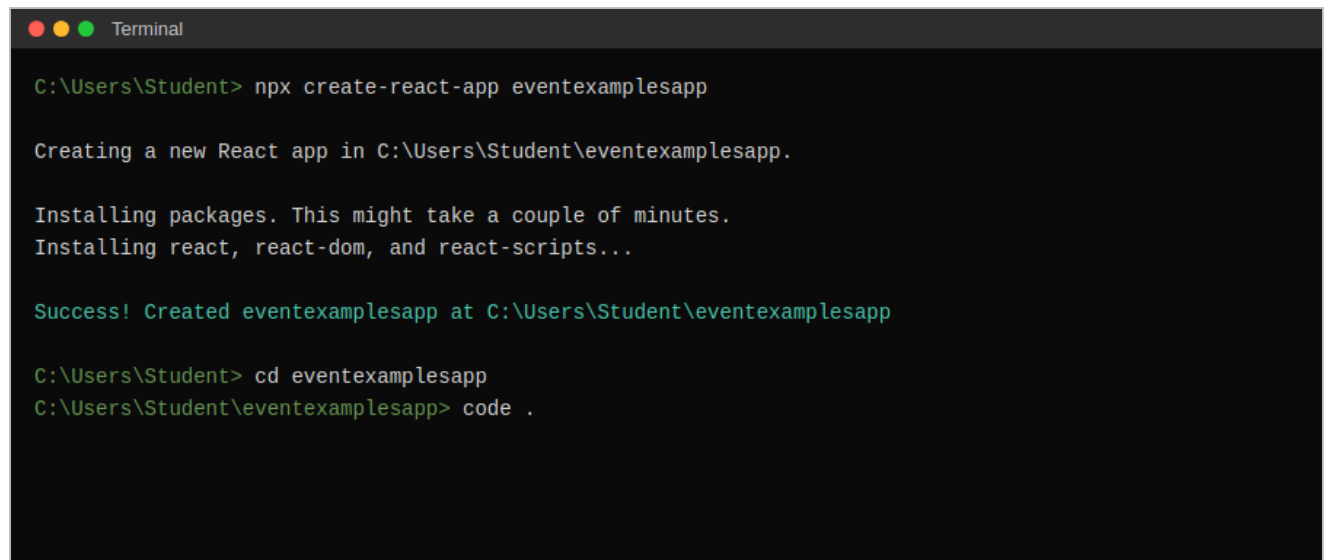
Objectives

- Explain React events
- Explain about event handlers
- Define Synthetic event
- Identify React event naming convention

In this lab I implemented event handling in React, used the "this" keyword, and used synthetic events.

Setup: Create the React app "eventexamplesapp"

Command used: `npx create-react-app eventexamplesapp`

A terminal window with a dark background and light green text. The title bar shows three colored circles (red, yellow, green) and the word "Terminal". The terminal output shows the command to create a new React app, the progress of installing packages, and the final success message. The user then navigates to the new directory and opens the code editor.

```
C:\Users\Student> npx create-react-app eventexamplesapp

Creating a new React app in C:\Users\Student\eventexamplesapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts...

Success! Created eventexamplesapp at C:\Users\Student\eventexamplesapp

C:\Users\Student> cd eventexamplesapp
C:\Users\Student\eventexamplesapp> code .
```

Question 1: Increment / Decrement buttons

The Increment button invokes two methods – one to increase the counter value and another to say Hello with a static message. The Decrement button decreases the counter value.

```
App.js
src > App.js

1 2 3 4 5 6 7 8 class App extends Component {
9 10 11 12 13   constructor(props) {
14 15 16 17 18     super(props);
19 20 21 22 23     this.state = { count: 0 };
24 25 26 27 28   }

    class="cmt">// a. increments the counter value
    incrementCount = () => {
      this.setState({ count: this.state.count + 1 });
    }

    class="cmt">// b. says Hello followed by a static message
    sayHello = () => {
      alert(class="str">"Hello! Counter has been incremented.");
    }

    class="cmt">// Increment button invokes both the above methods
    handleIncrement = () => {
      this.incrementCount();
      this.sayHello();
    }
  }
}
```

Question 2: "Say Welcome" button with an argument

The button invokes a function which takes "welcome" as an argument.

```
App.js
src > App.js

1   class="cmt">// function which takes class="str">"welcome" as an argument
2   sayWelcome = (message) => {
3     alert(message);
4   }
5
6   ...
7   <class="tag">button onClick={() => this.sayWelcome(class="str">"welcome")}>Say Welcome</class="ta
```

Question 3: Synthetic event "OnPress"

The button invokes a synthetic event handler which displays "I was clicked".

```
App.js
src > App.js

1 2 3 4 5   class="cmt">// synthetic event handler - class="str">"e" here is React's SyntheticEvent
6 7 8   handlePress = (e) => {
          alert(class="str">"I was clicked");
          console.log(e);
        }

    ...
    <class="tag">button onClick={this.handlePress}>OnPress</class="tag">button>
```

Question 4: CurrencyConvertor component

Converts Indian Rupees to Euro when the Convert button is clicked. The Click event invokes handleSubmit which performs the conversion.

CurrencyConvertor.js

src > CurrencyConvertor.js

```
1  class="kw">import React, { Component } from class="str">"str">'react';
2
3  class CurrencyConvertor extends Component {
4    constructor(props) {
5      super(props);
6      this.state = {
7        rupees: class="str">"str">'',
8        euro: null
9      };
10   }
11
12   handleChange = (e) => {
13     this.setState({ rupees: e.target.value });
14   }
15
16   handleSubmit = (e) => {
17     e.preventDefault();
18     const euroValue = this.state.rupees * 0.011;
19     this.setState({ euro: euroValue.toFixed(2) });
20   }
21
```

Complete App.js

```
App.js
src > App.js

1  class="kw">import React, { Component } from class="str">"str">'react';
2  class="kw">import CurrencyConvertor from class="str">"str">'./CurrencyConvertor';
3
4  class App extends Component {
5    constructor(props) {
6      super(props);
7      this.state = {
8        count: 0
9      };
10   }
11
12   class="cmt">// a. increments the counter value
13   incrementCount = () => {
14     this.setState({ count: this.state.count + 1 });
15   }
16
17   class="cmt">// b. says Hello followed by a static message
18   sayHello = () => {
19     alert(class="str">"Hello! Counter has been incremented.");
20   }
21
22   class="cmt">// Increment button invokes both the above methods
23   handleIncrement = () => {
24     this.incrementCount();
25     this.sayHello();
26   }
27
28   decrementCount = () => {
29     this.setState({ count: this.state.count - 1 });
30   }
31
32   class="cmt">// function which takes class="str">"welcome" as an argument
33   sayWelcome = (message) => {
34     alert(message);
35   }
36
37   class="cmt">// synthetic event handler - class="str">"e" here is React's SyntheticEvent
```

Build and run using npm start

```
Terminal

C:\Users\Student\eventexamplesapp> npm start

Compiling...

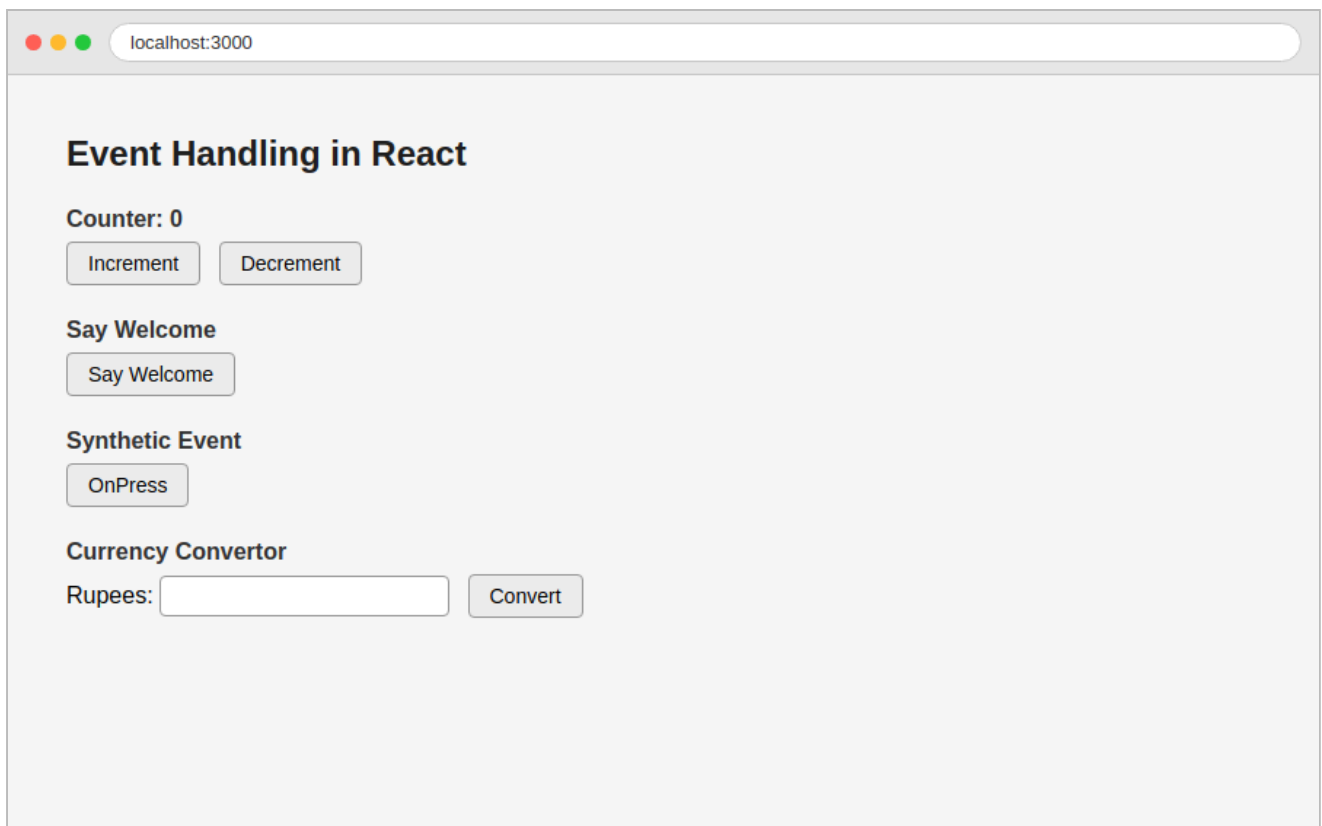
Compiled successfully!

You can now view eventexamplesapp in the browser.

Local:      http://localhost:3000
On Your Network:  http://192.168.1.7:3000

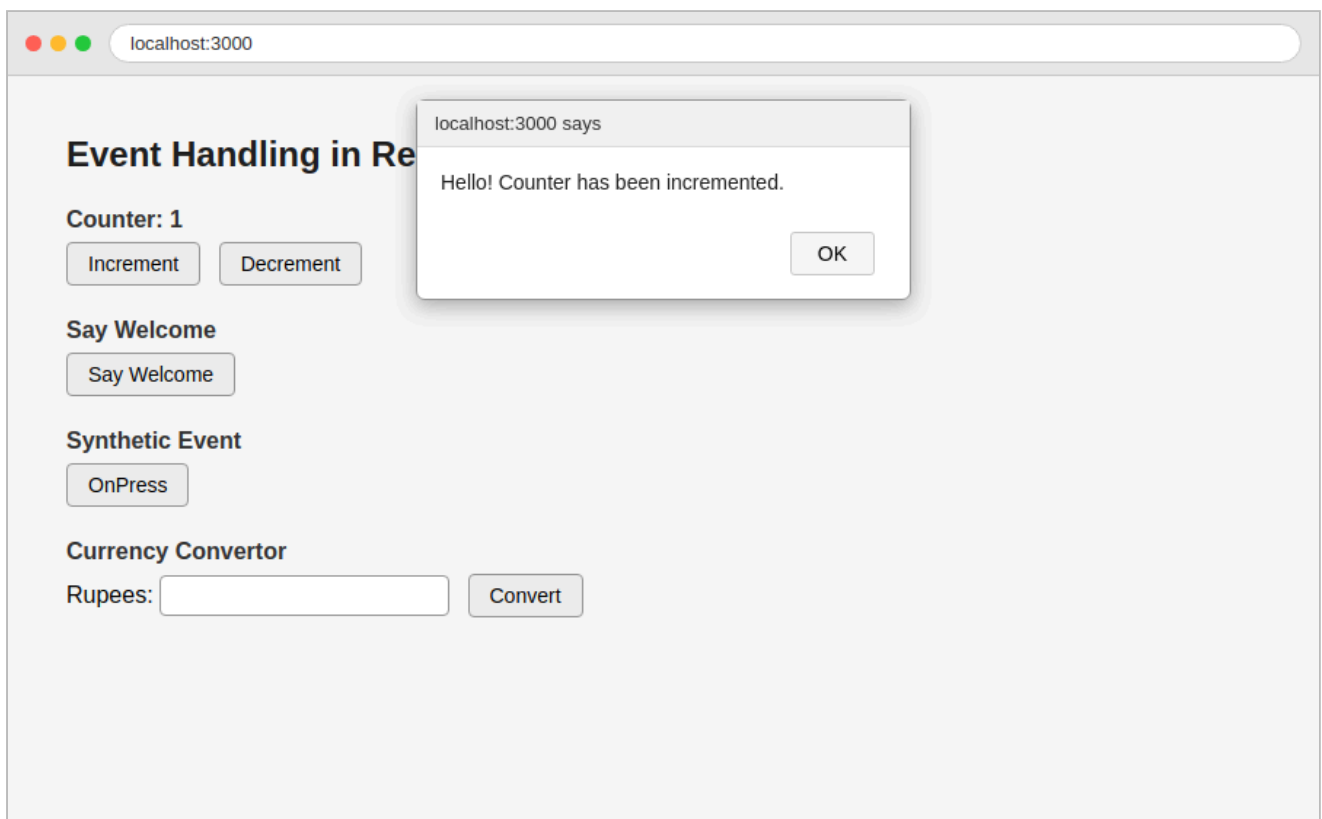
webpack compiled successfully
```

Output: Initial page

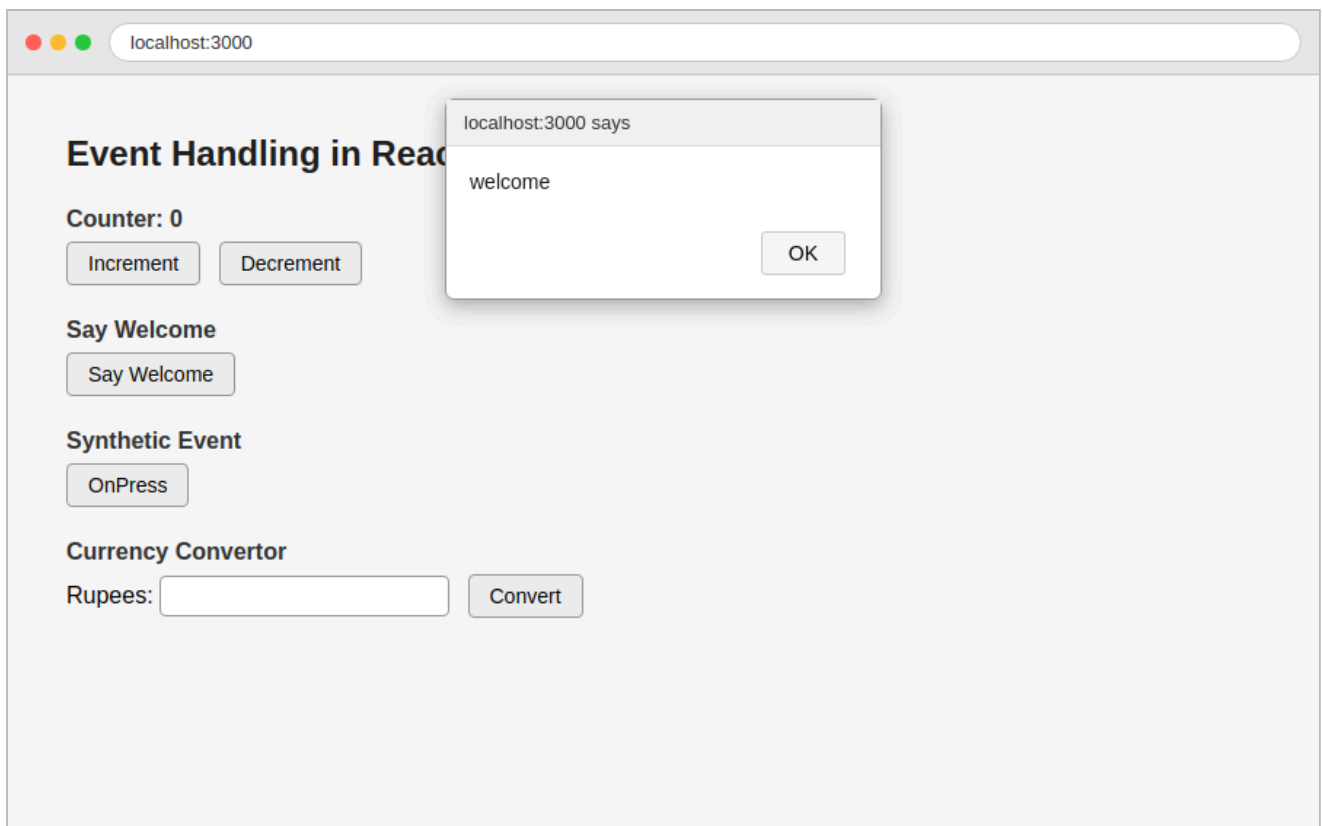


Output: Increment button clicked

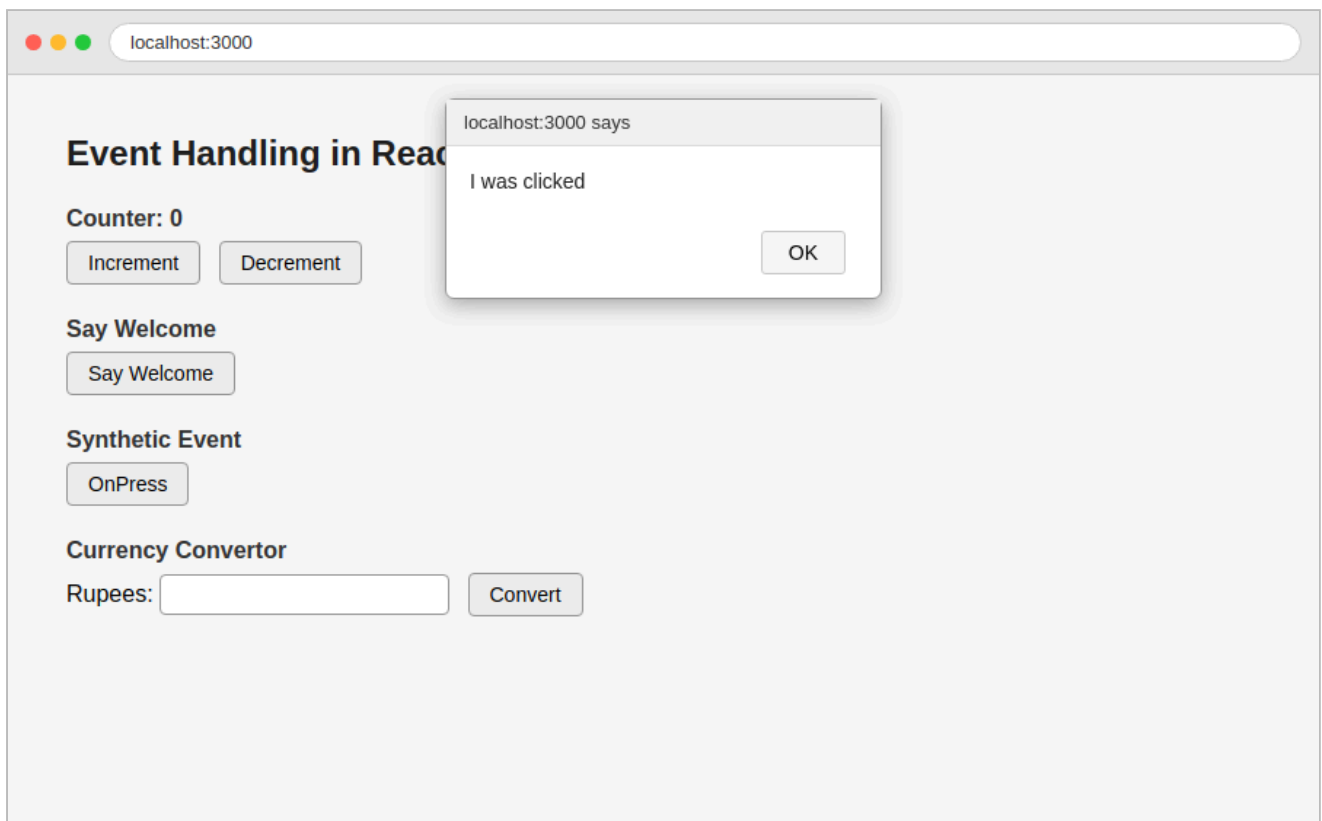
Counter value increases and the Hello alert message is shown.



Output: Say Welcome button clicked

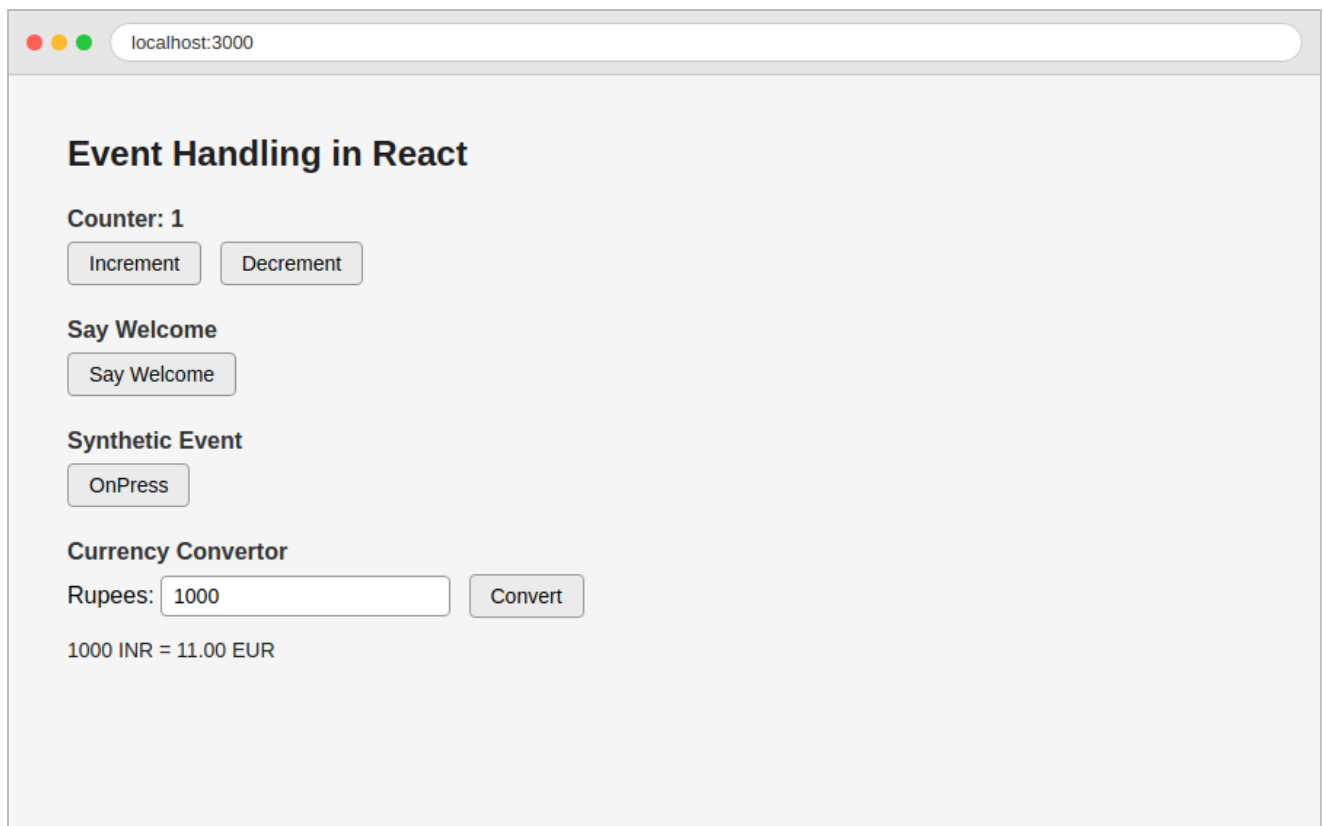


Output: OnPress button clicked (Synthetic Event)



Output: Currency Convertor result

1000 Rupees converted to Euro on clicking Convert.



Result

Event handling in React was implemented successfully using onClick handlers, the "this" keyword to bind component methods, passing arguments to event handlers, and React's synthetic event system. The CurrencyConvertor component was also implemented and tested.